# Which route would you choose?

You need to move 15 pallets of fragile electronics from Fremont to Austin. Your budget is $4,000.

### OPTION A / LOWER COST

**$2,100**

Fremont > Phoenix > Austin

One transfer

### OPTION B / EARLIER ARRIVAL

**$3,400**

Fremont > Austin, direct

Arrives 8 hours earlier

## Before choosing, what else would you need to know?

Illustrative prices and services from the fixed synthetic scenario.

# Meet RouteMind AI

A conversational assistant for comparing suitable ground-freight routes.

DESCRIBE

## Tell us the shipment

Locations, quantity, budget, deadline and cargo-handling needs.

COMPARE

## Understand the options

Review costs, arrival times, transfers and supporting information.

EXPLORE

## Ask what changes

“Make it faster, but keep the budget.” Then review the trade-off.

**Decision support: the user reviews the options and makes the final choice.**

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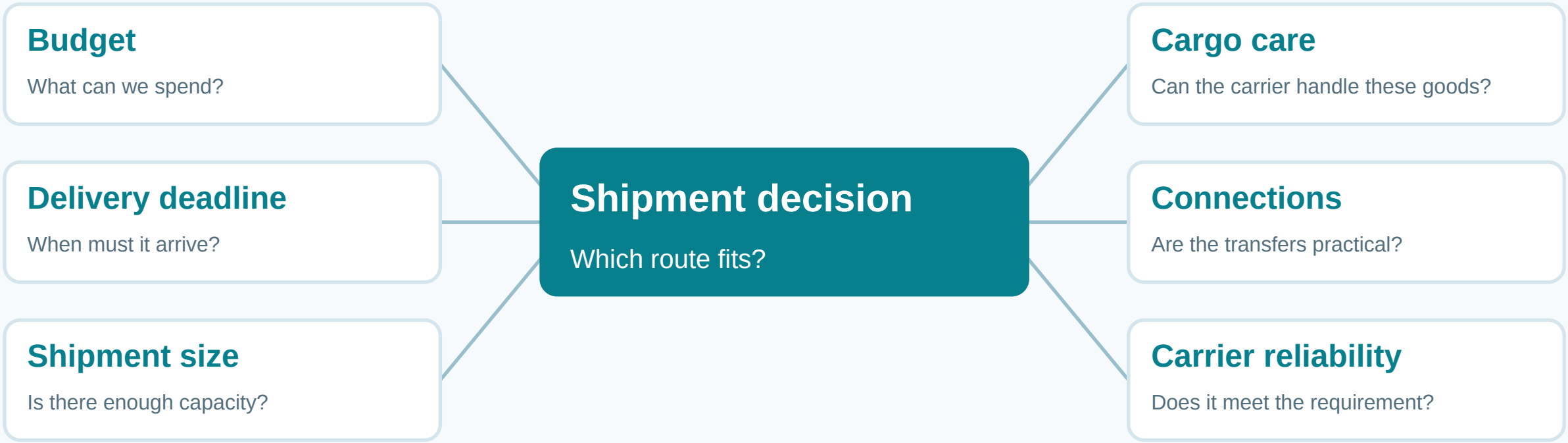
# What makes a route suitable?

A route must satisfy the shipment’s requirements—not just offer a low price.




**A change to one requirement can change the best route.**

# From a request to a decision you can explain

The challenge is to connect the user’s intent, operational constraints and supporting information.

**How can we turn a natural-language shipment request into feasible, explainable route options while keeping operational facts separate from document-based interpretation?**

UNDERSTAND

## Clarify the need

Ask about missing or ambiguous requirements before planning.

VERIFY

## Check what is feasible

Respect budget, timing, capacity, connections and handling requirements.

EXPLAIN

## Make evidence visible

Show the trade-offs and say when information is missing or conflicting.

# Two kinds of information. Two different jobs.

SQL verifies operational facts; RAG retrieves supporting document passages.

STRUCTURED FACTS / SQL + OPTIMIZER

## What services actually fit?

Prices, schedules, available pallet space and explicit handling flags.

Query the facts, then calculate and verify feasible routes.

WRITTEN INFORMATION / RAG

## What do the documents say?

Carrier descriptions and handling guidance, expressed in varied language.

Find relevant passages and show their sources.

**RAG supports document understanding.**

**It does not replace database checks or determine route eligibility.**